Nabeel Shan

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Education

National University of Sciences and Technology (NUST) **Islamabad, Pakistan**
Bachelor of Engineering in Software Engineering (CGPA: 3.52/4.00)

Experience

NLP & LLM Research Intern **Jun 2025 – Aug 2025**
The Centre of Excellence for Technology Quantum and Artificial Intelligence Pakistan (CETQAP) *Islamabad, PK*

- Applied **LLM fine-tuning and prompt engineering** to automate summarization of technical reports in quantum computing, accelerating literature review by **25–30%**.
- Developed and curated a dataset of **500+ adversarial prompts** to benchmark 3 commercial LLMs, uncovering key weaknesses in instruction-following and guiding model improvement.
- Authored research deliverables by synthesizing literature and documenting experiments, strengthening communication of AI project outcomes to senior researchers.

AI/ML Intern **Apr 2025 – May 2025**
Software Productivity Strategists (SPS) *Remote – Islamabad, Pakistan*

- Integrated a fine-tuned **RoBERTa** model into a sentiment analysis API (Python, Flask), collaborating in an Agile team using Git workflows.
- Enhanced NLP feature by fine-tuning **BERT** on client data, boosting NER accuracy to **92%**.
- Built a reusable preprocessing pipeline with Pandas, cutting manual cleaning time by **30%** and accelerating experiments.

AI & Machine Learning Projects

LLM Alignment in Practice: Building a Complete RLHF Pipeline from Scratch

- Established a high-performing baseline by engineering a Supervised Fine-Tuning (SFT) stage, improving GPT-2's ROUGE-1 score by **97%** while benchmarking PEFT (QLoRA) against full-tuning.
- Constructed a high-fidelity reward model by fine-tuning on 33k+ preference pairs, achieving **∼98% accuracy** in predicting human judgments to create a robust reward signal for PPO.
- Implemented the final alignment pipeline using Proximal Policy Optimization (PPO), steering the model to achieve a **54% higher reward score** than the SFT baseline.

Efficient and Safe Adaptation of LLMs via PEFT and RLHF

- Architected an end-to-end adaptation pipeline for **FLAN-T5**, systematically benchmarking SFT vs. PEFT and achieving an **81%** relative improvement in summarization quality (ROUGE-1: 0.233 → 0.422).
- Optimized fine-tuning with **LoRA**, reaching **97%** of full SFT performance while training only **1.4%** of parameters.
- Aligned model behavior for AI safety using **RLHF** (PPO), cutting toxic content by **9.2%** using a RoBERTa-based reward model.

Parameter-Efficient Fine-Tuning of Transformers: An Empirical Study

- Systematically benchmarked 5 adaptation strategies in an empirical study on IMDB sentiment analysis, demonstrating that PEFT (Adapters) achieves performance parity with full fine-tuning (**85.6%** vs. **86.0%**) while requiring **13**× fewer trainable parameters.
- Engineered key PEFT methods (**Adapters, LoRA**) from first principles in **PyTorch** to validate their efficiency, reducing a target layer's trainable parameters by over **96%** (456 vs. 12,800).

GPT-Forge: From-Scratch Transformer for Text Generation

- Built a decoder-only **Transformer** in PyTorch with causal self-attention for autoregressive text generation.
- Scaled model capacity from **22M** to **52M** parameters via a reproducible GPU-accelerated training pipeline.
- Achieved a **2.4%** perplexity reduction after scaling, validating the from-scratch architecture's effective learning capability.

Empirical Analysis of RAG via Hybrid Search

- Proved that a hybrid search pipeline using Reciprocal Rank Fusion (RRF) improves **RAG** response grounding and coherence by **20–30%**, addressing the key challenge of suboptimal context retrieval in single-retriever systems.
- Engineered a reproducible benchmarking framework to systematically A/B test retrieval methods, integrating keyword (BM25), semantic, and RRF strategies.

Deep Learning from Scratch in NumPy: DNN, CNN & RNN

- Built fully-vectorized DNN, CNN, and RNN in NumPy, implementing layers and BPTT from scratch.
- Implemented advanced training dynamics **(Adam, RMSProp, Momentum)**, regularization (Dropout, L2), and initializers (He/Xavier) to stabilize convergence.
- Validated with real-world tasks: CNN on Sign Language Digits (image classification) and RNN for text generation, analyzing loss curves and hyperparameters.

Technical Skills

Languages: Python, C++, SQL
Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face, LangChain, Scikit-learn, NumPy, Pandas, Matplotlib
Core AI/ML Expertise:

- **LLMs & NLP:** Transformers Architectures (BERT, GPT, T5), PEFT (LoRA, QLoRA, Adapters), Knowledge Distillation, SFT, RAG, Summarization, NER, Translation
- **LLM Alignment & Safety:** RLHF, PPO, DPO, Reward Modeling & Hacking Mitigation
- **Computer Vision, Speech & Multimodal:** ViT, VLMs, U-Net, YOLO, ResNet; Projects: Semantic Segmentation, Face Recognition, Wake Word Detection.

MLOps & Tools: Git, Docker, Weights & Biases, Linux/Bash, Hugging Face Hub/Spaces

Certifications & Specializations

DeepLearning.AI / Stanford University — via Coursera

- Machine Learning Specialization
- Deep Learning Specialization
- Natural Language Processing Specialization
- TensorFlow Developer Specialization

IBM — via Coursera

- AI Engineering Professional Certificate
- Generative AI Engineering with LLMs Specialization
- IBM Deep Learning with PyTorch, Keras and Tensorflow Specialization.

Leadership & Community Involvement

- **Organizing Committee Member, Devcon Tech Competition** — Software Society, MCS NUST
  Coordinated logistics and judging for the AI/ML track of the university's flagship tech competition, engaging over 200 participants.
- **Active Member** — Community Service Society, MCS NUST
  Contributed to numerous social welfare initiatives, including donation drives, plantation campaigns, and blood donation events.